

TITLE: Prediction of Microplastic Concentration in Marine Water Using Machine Learning

ABSTRACT:

This project aims to predict microplastic concentration in marine water using supervised machine learning regression models. The dataset used is the NOAA Marine Microplastics Database, which contains tabular information about sampling locations, water depths, marine settings, mesh size, and sampling methods. The goal is to build a regression model that estimates microplastic concentration based on geographic, environmental, and sampling-related variables. The model helps streamline pollution assessment and support environmental monitoring.

PROBLEM STATEMENT:

Traditional laboratory-based measurement of microplastic concentration is time-consuming and expensive. Researchers need a faster and scalable way to estimate pollution levels across the ocean. This project develops a machine learning system that predicts microplastic concentration using metadata features provided in the NOAA dataset. The model captures spatial and sampling-related patterns to estimate pollution levels efficiently.

DATASET DESCRIPTION:

Dataset Source: NOAA Marine Microplastics Database

Type: Tabular (CSV)

Target Variable (Output): Microplastics measurement (continuous numeric concentration)

Input Features (Independent Variables):

- Geographic Features: Latitude, Longitude, Region, Ocean, Subregion, Country, State
- Environmental Features: Marine Setting, Ocean Bottom Depth (m), Water Sample Depth (m), Sediment Sample Depth (m)
- Sampling Method Features: Sampling Method, Mesh size (mm), Transect No, Volunteers Number, Collecting Time (min)
- Metadata: Date (MM-DD-YYYY), Short Reference, ORGANIZATION

ML APPROACH:

Type: Supervised Learning

Task: Regression

Models Used:

- Linear Regression
- Ridge Regression
- Lasso Regression

- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor (recommended)

Evaluation Metrics:

- MAE
- MSE
- RMSE
- R^2 Score

KEY POINTS:

- Microplastic concentration is influenced by sampling location & method.
- NOAA dataset does not include chemical water parameters; hence the model works on geographic & sampling metadata.
- This is a valid ML regression problem as the target is a continuous variable.
- Tree-based models handle non-linearity and perform best.

CONCLUSION:

The project demonstrates that machine learning can be effectively used to estimate microplastic concentration using metadata alone. The model can assist marine researchers and environmental agencies in rapid pollution monitoring.